SAPPHIRE PROGRAMMING LANGUAGE

Complete Coding Manual, Syntax Reference, ML & Agent Library Guide

1. Introduction to Sapphire Language

Sapphire is a modern high-level programming language engineered specifically for **PC Automation**, **Deep Learning & Neural Training**, **AI Agent Architectures**, and **Colorless Parallel Concurrency**. Sapphire combines clean readable syntax with zero-boilerplate standard library primitives for tensors, autograd, GPU acceleration, OS manipulation, and LLM reasoning.

Core Pipeline: Data to Autonomous Execution

Data → Training → Model → Reasoning → Memory → Planning → Tool use → Autonomous execution

2. Language Syntax & Fundamentals (.sp)

```
let agent_name = "Sapphire-Alpha";
let version = 1.0;
let is_active = true;
let tensor_data = ml.tensor([[1, 2], [3, 4]]);

print("Agent {agent_name} running Sapphire v{version}");
```

3. Deep Learning & AI Standard Libraries

ml Module

Tensors, autograd, datasets, model architectures, distributed training, numerical kernels, and GPU/TPU acceleration.

```
let model = ml.model.mlp([16, 64, 2], 'relu');
let result = ml.train.fit(model, dataset, ml.loss.mse, ml.optim.adam(0.01), 5, 32, 2);
```

ai Module

Native LLM reasoning with local Ollama inference and automatic Groq API cloud fallback.

```
let opinion = ai.prompt('Analyze RAM load and suggest optimization.');
print('AI Opinion: {opinion}');
```

agent Module

Short & long-term memory, goal planning, tool registration, permission policies, and autonomy loop.

```
agent.memory.remember('accuracy', 0.95);
agent.tools.register('deploy', 'Deploys model', fn() { os.notify('Alert', 'Deployed'); });
agent.autonomy.run_loop('Deploy model v1', 5);
```